Network Management: Comprehensive Study Guide

1. Definitions of Network Management

Network management can refer to anything from everyday operational activities to formal, structured frameworks for keeping networks and their services working properly.

Informal Definition

Informally, **network management** means the activities involved in running a network and the technology used to support those activities. A major part of network management is **constant monitoring**, which gives administrators a real-time understanding of what is happening on the network.

Formal Definition: OAMP

A formal definition describes network management as the activities, methods, procedures, and tools used for the **Operation, Administration, Maintenance, and Provisioning (OAMP)** of networked systems.

- **Operation** - Keeps the network and its services running smoothly. It includes continuous monitoring so problems can be identified as early as possible.
- **Administration** - Manages network resources and determines how those resources are assigned.
- **Maintenance** - Includes repairs, upgrades, and proactive improvements that keep the network reliable.
- **Provisioning** - Configures network resources so they can support specific services.

Narrower vs. Broad Management Domains

Although the term network management is often used broadly, it can be divided into several more specific areas:

- **Network Management:** Management of the communication network itself.
- **Systems Management:** Management of end systems connected to the network, such as Linux, Windows, or Unix servers.
- **Application Management:** Management of software applications running on those systems, such as finance, database, or web applications.
- **Service Management:** An umbrella concept used when networks, systems, and applications work together to provide a specific service.

2. Two Analogies for Network Management

Two useful analogies help explain the complexity and requirements of managing a network.

The Health Care Analogy: The Network as a Patient

This analogy compares a network to a living organism - specifically, an ICU patient that requires constant observation and care.

- **The Pulse:** Critical functions of network equipment must be monitored continuously. A malfunction in these functions should trigger an immediate alarm.
- **Vitals (Temperature, Blood Pressure):** Network trends, such as an increasing rate of dropped packets or link utilization approaching 100%, can signal developing capacity problems. These are monitored regularly, but not as frequently as the network's 'pulse.'

- **MRI/Blood Samples:** Specialized diagnostic or troubleshooting procedures are used in unusual situations, such as taking a device offline for testing.
- **Diagnosis and Treatment:** Network managers act like doctors. They adjust or 'tune' parameters and configurations, then monitor the response to confirm the diagnosis or change the treatment plan.

The Party Analogy: Running an Event

Running a network shares several characteristics with planning and executing a major event.

- **Planning:** Determine required services, capacity, and network topology - for example, deciding between many small branch offices and one large campus.
- **Setup:** Commission equipment, tune configuration parameters, and manage technical dependencies across different equipment types and versions.
- **Continuous Nature:** Unlike a party, a network does not have a clear ending or cleanup period. Adjustments, decommissioning, and replacements must happen seamlessly while end users continue using the network.

3. Why Network Management Matters to the Business

Network management is a critical factor in the economic success of organizations that depend on communication services.

Total Cost of Ownership (TCO)

The total cost of network equipment can be divided into two major categories:

1. **Equipment Cost:** The purchase price of the hardware itself, spread over the equipment's useful lifetime.
2. **Operational Cost:** Ongoing expenses such as personnel, electricity, physical space, and operations-support infrastructure. Effective management tools make operations more efficient and can therefore reduce TCO.

Revenue Generation and Turn-up Speed

Effective network management can directly affect revenue:

- **Turn-up Speed:** Efficient service provisioning reduces the time between a customer ordering a service and that service becoming active. Faster turn-up means revenue begins sooner and customers are less likely to go to a competitor.
- **Value-Added Features:** Management capabilities such as online accounting or charge tracking and web-based feature configuration can make services more attractive to customers.

Quality and Reliability: Five Nines

Carrier-class networks may be engineered for **"Five Nines" availability: 99.999% availability.**

- **Redundancy:** Use redundant hardware and redundant communication paths so one failure does not take down the entire service.
- **Fault Management:** Intelligent systems can automatically reroute traffic around fiber cuts or hardware failures.
- **Service Level Agreements (SLAs):** Management systems help organizations meet guaranteed quality-of-service requirements and other agreed performance metrics.

Compliance

Network management can also be driven by legal and regulatory requirements. One example is the **Sarbanes-Oxley Act (SOX)**, enacted after major accounting scandals such as Enron and WorldCom. SOX requires enterprise IT departments to maintain strict controls and oversight of their data and communication systems.

4. The Network Management Industry

The network-management industry includes different groups with distinct interests, responsibilities, and perspectives.

Users of Network Management

- **Service Providers:** Organizations such as telecommunications carriers and ISPs that earn revenue by operating networks. Network management helps protect revenue and supports competition in a commoditized market.
- **Enterprise IT Departments:** Often function as 'mini service providers' inside their own organizations. They are usually cost centers rather than direct revenue generators and may outsource operations to **Managed Service Providers**.
- **End Users/Operators:** The network administrators, technicians, planners, and other personnel who perform the day-to-day work required to keep the network operating.

Providers of Network Management

- **Equipment Vendors:** Primarily sell networking hardware. They often include basic management software with their equipment and may use unique management features to differentiate their products.
- **Third-Party Application/Management Vendors:** Fill the 'management gap' with vendor-independent software that supports multi-vendor environments and focuses on operational workflows rather than one specific type of hardware.
- **Systems Integrators:** Integrate different management applications into a specific network and operations environment. They fill functional gaps and create customized interface adaptations.

5. Challenges of Network Management

As network complexity increases, network management becomes more important - and more difficult.

Technical Challenges

- **Technical Dependencies:** Managers must work with many equipment types, technologies, and software versions.
- **Automation vs. Human Error:** Misconfiguration is a major cause of reduced network availability. Automated end-to-end provisioning can improve efficiency and reduce errors caused by manual configuration.
- **Scalability:** Large service providers may need an 'army' of personnel and sophisticated management tools, while a smaller organization may rely on one person to handle many network-management tasks.

Integration and Business Challenges

- **The Multi-vendor Reality:** Organizations often discover that no single management tool can handle their entire infrastructure. This creates a need for systems that can work from shared information, such as a unified network inventory.
- **Commoditization:** Services such as telephone, internet, and streaming are increasingly treated as commodities. Service providers compete by lowering operating costs, introducing services quickly, and

maintaining high quality.

- **Opportunity Cost:** Failing to meet quality standards can cause lost revenue, inefficient use of resources, and higher operational costs spent fixing preventable problems.

Quick Recall

OAMP = Operation, Administration, Maintenance, Provisioning.
Five Nines = 99.999% availability.
TCO = equipment cost + operational cost.
Automation helps reduce configuration errors.
Multi-vendor networks create integration challenges.